

Poster: Wearable Input Device for Smart Glasses Based on a Wristband-Type Motion-Aware Touch Panel

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ABSTRACT

This paper presents a smart glasses-augmented, wearable input device based on a wristband-type, motion-aware touch panel. With this novel input device, a user can manipulate the smart glasses' augmented reality (AR) system by selecting and moving contents via touching and dragging, as well as scrolling, navigating, and controlling 3D objects by rotating the wrist. The proposed device can be used in an AR environment to select and manipulate virtual 2D and 3D content. The proposed interface offers the following advantages: 1) the touch screen panel helps users to control the smart glasses' AR system accurately, without any limitations of gesture and speech recognition systems; environmental noise and distortion, and multi-leveled pattern recognition task; 2) the motion of the wrist increases the Degrees of Freedom (DOF) of the system – the combination with 2 DOF of touch interaction allows accurate 3 DOF interaction and various ways of interacting, enabling users to carry out tasks via a Head-Mounted Display (HMD) quickly and intuitively; and 3) the simple hardware composition allows the device to be easily embedded into existing devices such as smart watches.

Keywords: input devices, wearable, multimodal interaction, 3D interaction, HMD, smart glasses, wristband-type, motion-aware, touch-aware

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, augmented and virtual realities H.5.2 [Information Interfaces and Presentation]: User Interfaces—Input devices and strategies

1 INTRODUCTION

In the electronics market, smart devices have advanced rapidly. Following smart phones, the current electronics market is focusing research on smart glasses and related applications.

Smart glasses are simply reinvented wearable computing for a new era. However, smart glasses are not yet a perfect tool for human-computer interaction or wearable computing, due to the limitations of its input system, and speech and gesture recognition. Gesture recognition technology still fails to identify contexts due to the equipment used, image noise, and speech recognition technology still has difficulty identifying vocabularies due to environmental noise, acoustic distortions, and load for multi-leveled pattern recognition tasks [1][2]. On the other hand, although smart watches have also received attention from the market, they are criticized due to application limitations.

In this research, we suggest a wristband-type 3D input device

for smart glasses to expand their availability and capability by using combinations of motion-aware and touch panel systems. The wristband-type smart device would assist smart glasses users in manipulating various contents delicately.

2 DESIGN

The design consists of motion-aware and touch-panel parts to provide a natural and accurate interface. The device is a wearable input interface for smart glasses that uses the wrist and fingers – those body parts which humans use the most actively and freely; the DOF of the wrists and their range of motion are the highest [3].

2.1 Hardware Composite

The device consists of two parts, a Touch Screen Panel (TSP) and an Inertial Measurement Unit (IMU). The TSP is attached on top of a wristband and the IMU is positioned at the bottom of the wristband, as shown in Figure 1.

The TSP is fabricated by combining Indium Tin Oxide (ITO) and a multi-layer composite of silver-nanowires and carbon-nanotubes as electrodes (30x50 mm). Each side of the combined electrodes are connected to a four-point resistance measure unit [4]; finger-touch is applied to move the cursor on the screen of a laptop, as well as to write text and draw shapes.

The IMU is a Sparkfun 9DOF sensor stick (35x11x2.4 mm) [5]. The values from an accelerometer, magnetometer, and gyroscope sensors are fused by the razor-9dof-ahrs method [6].

Finally, these units are integrated on a transparent and flexible band, and interwork 3D HMD glasses with the wristband device

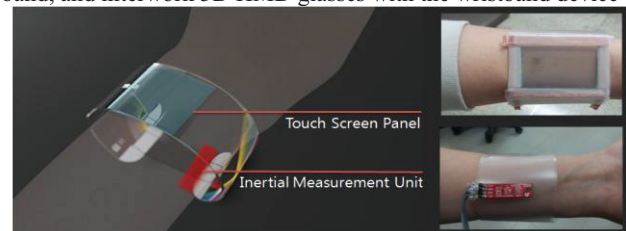


Figure 1. Concept sketch and actual implementation by transparent TSP and IMU on a transparent wristband

2.2 Interaction

The user interface is composed of a 3D HMD and the motion-aware touch panel. In the HMD-wearing environment, the user controls the menus, augmented information, and augmented objects by the motion-aware touch panel. First, the user see the display of HMD and move the cursor by dragging their finger on TSP with the cursor on the display. The finger's short touch is recognized as a click of a mouse input. The TSP is used to point or select, like the touchpad of a laptop. Second, IMU is used to detect motion of the wrist. The user can rotate the wrist to manipulate systems of the wearable computer. The axes of rotation are combined differently for applications in 1, 2, and 3-dimensional interactions.

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3 EXPERIMENT

3.1 Method and Process

We hired 3 subjects from our university (2 males and 1 female, average age 26). There is an independent variable, DOF, which is the number of axes for which the rotation of the wrist is detected. The goal of this experiment is to estimate the precision and task completion time of using the wristband-type touch panel, depending on the DOF given to be controlled by wrist rotation.

The participants rotated their wrist to the given angles of yaw, pitch, and roll, and then to click on the screen with the TSP on the wrist. The angles were from 1 to 3 DOF. For 1 DOF, each yaw, pitch, and roll angle was given. For 2 DOF, two of three compositions (yaw, pitch, and roll) were given. For 3DOF, all of the three compositions were given together. When participants think that their wrist position corresponds to the given angles, they have to click the screen by pushing the surface of the touch panel on the wrist. To guide participants, both the given objects and the wrist position are displayed together.

3.2 Results and Discussion

We estimated the time and error to complete each trial. The result shows that task completion time increases as the number of DOF increases, as shown in Figure 3d. We additionally analyzed the task completion time within the same numbers of DOF, as shown in Figures 3a, 3b, and 3c. We also estimated errors of angles, the difference between the target angles and the angles at the moment when the user finishes a trial. The result shows that 1 DOF produces significantly fewer errors than 2 or 3 DOF cases. However, 2 DOF produced more errors than 3 DOF. According to these results, although higher DOF provide more variables to control, the experimental results illustrate that it is not more effective than a lower DOF in speed and accuracy. Therefore, the interaction should not have the highest DOF, but rather has to have the minimum DOF following the number of variables of each different control. As a result, the 1D, 2D, and 3D interactions use 2, 2, and 3 DOF of control, respectively.

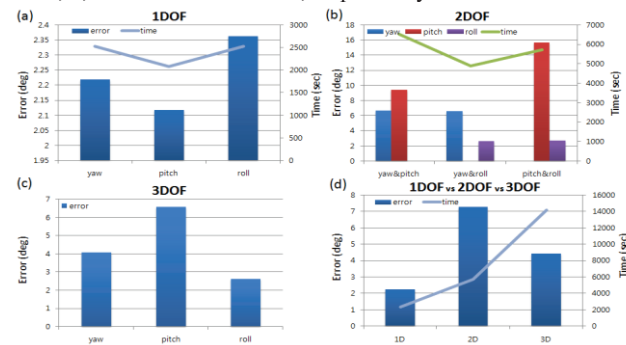


Figure 2. Precision and task completion time in each dimension of (a) 1 DOF, (b) 2 DOF, and (c) 3 DOF, as well as (d) overall task completion time and error for each DOF

4 APPLICATION

4.1 Content Switching and Navigation

The first application is 1D interaction; users can switch contents with the device. The different speed of rotation triggers the change of contents on the display, including the wallpaper, schedule, and email. The device provides personal interaction by using unremarkable movement of the wrist; smart glasses users feel embarrassed when they say something or move their body to change contents in the air, out of bystanders' fields of view.

The second application is 2D interaction; people can scroll the field of view. Rotating a user's wrist or using a TSP on the device leads to seeing the other side of the scene displayed on smart glasses. As the size of an HMD is small, the contents shown to a user at once are limited. This interaction can expand the range of contents and applications of smart glasses by extending the range of view a user can see and control.

4.2 Virtual Object Manipulation

The third application is 3D interaction; 3D virtual objects can be easily manipulated by the device. For selection, a user can click on the 3D object. The user can then use drag-and-drop for translation of the object by using the TSP. When the user wants to rotate the 3D virtual object, he or she can rotate the wrist, making the 3D object rotate according to the wrist rotation. This application suggests that the smart wristband is a 3D input device capable of controlling 3D virtual environments.

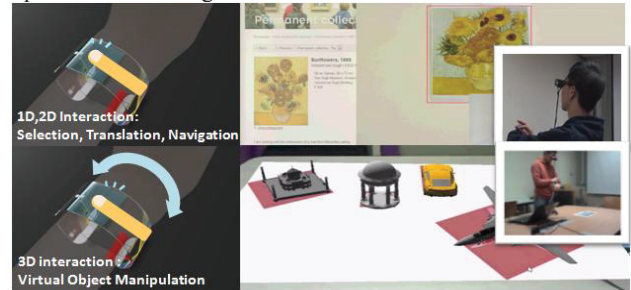


Figure 3. Applications for the smart wristband 3D input device: selection (1D), translation and navigation (2D), and virtual object manipulation (3D)

5 CONCLUSION

In this research, we present a novel 3D input device for smart glasses. The device consists of IMU and TSP using finger tactile and wrist rotation, while providing correct selection of 2D contents on screen and 3D objects in virtual reality to select and command correctly, directly, and easily. Through these experiments and applications, we observed users' behaviors with regard to how quickly and accurately they could utilize our device. The result shows that users were able to effectively use the device to interact with information on the HMD; they select and control contents correctly by using tactile input via a finger and by using wrist motions. The simple and transparent hardware composition of the device also provides a wide range of applications that interact with existing devices, such as smart watches, smart phones, and so on. Since it is equipped with a touch device, the 2D input device, it can also be used with existing technologies as well, like radios, smart TVs, and even computers.

To conclude, we developed a smart wristband interface for smart glasses, a smart device of the next generation.

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